Marks in ICP

>= ~~90~~ 95	9, 10
~~80~~ 85 to ~~89~~ 94	8
~~70~~ 75 to ~~79~~ 84	7
~~60~~ 65 to ~~69~~ 74	6
50 to 59	5
40 to 49	4
< 40	default

icpmark / 10

78 / 10 = 7

100 / 10 = 10

23 / 10 = 2

Grades

O

A

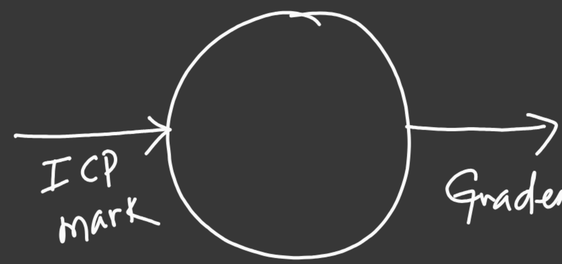
B

C

D

E

F



```
if(icpmark >= 90)
{
  s.o.pln("O");
}
else if( icpmark >= 80 && icpmark < 90)
{
  s.o.pln("A");
}
else if (icpmark >=70 && icpmark <80)
{
  s.o.pln("B");
}
.....
else
{
  s.o.pln("F");
}
```

85 >=90

85 >= 80

85 <90

Switch-Case :- Alternative use of if-else if ladder.

- From multiple option it will choose one.

Syntax

```
switch (Switch_Variable)
{
    case Case_Value:
        Stmts
        break;

    Case Case_Value:
        Stmts
        break;

    .
    .
    .

    (optional) default:
        Stmts
}
```

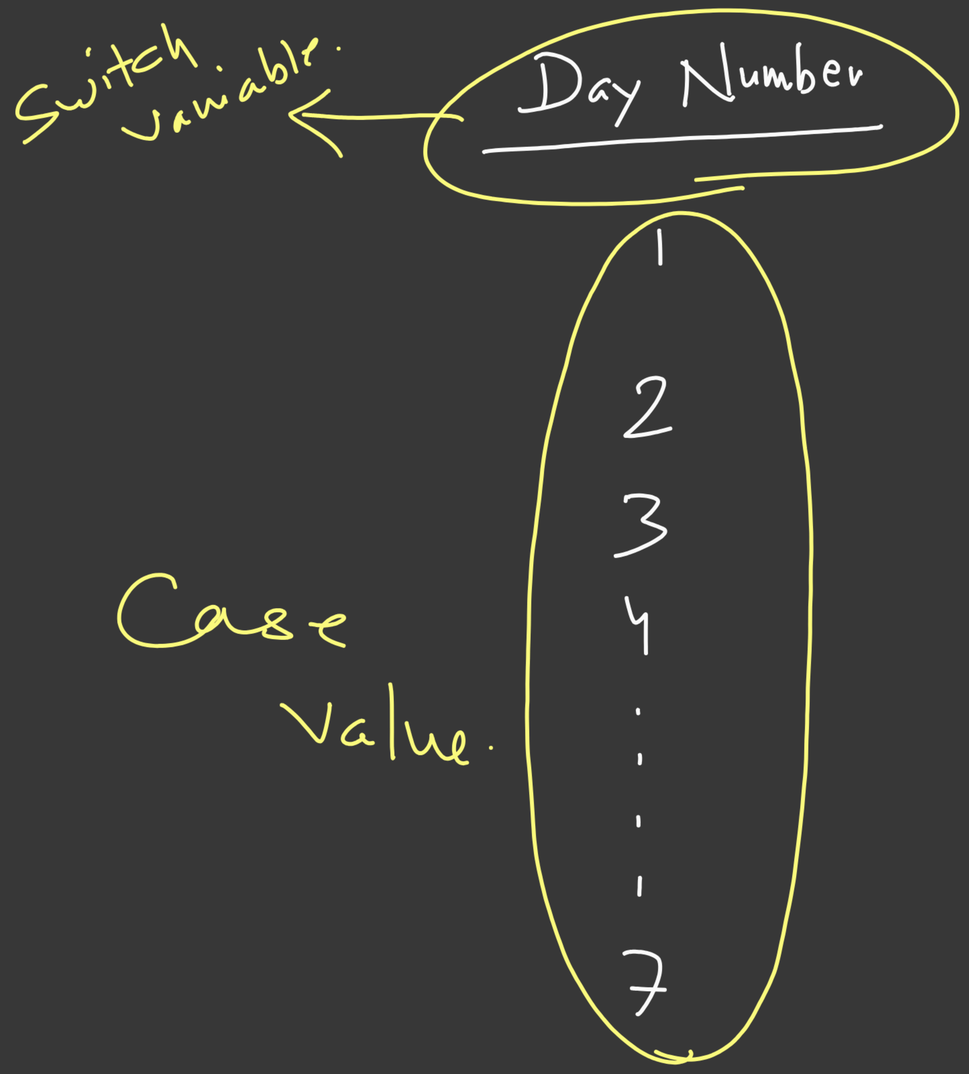
Keywords

switch
Case
break
default

To Come outside the switch.

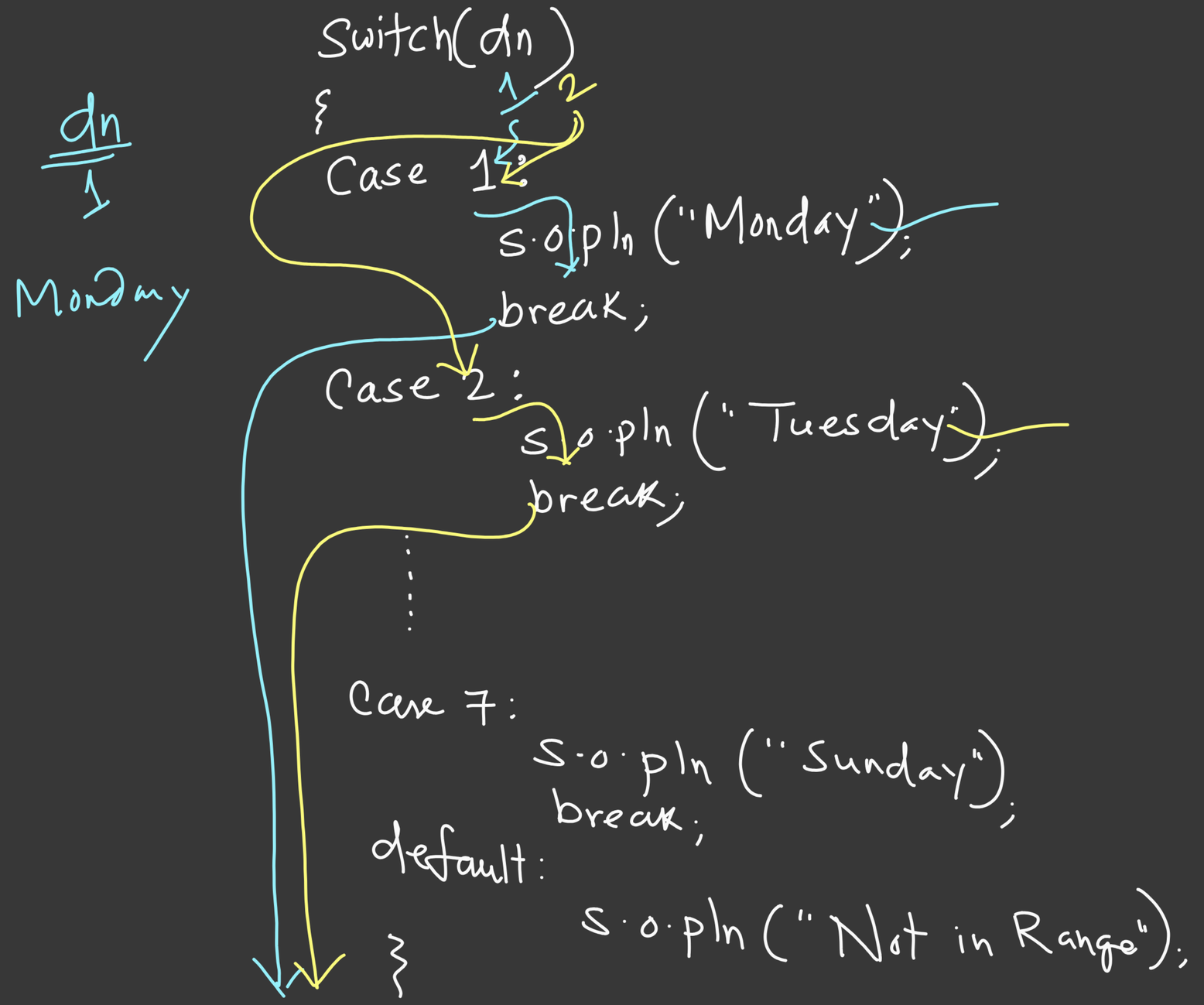
Most important two things

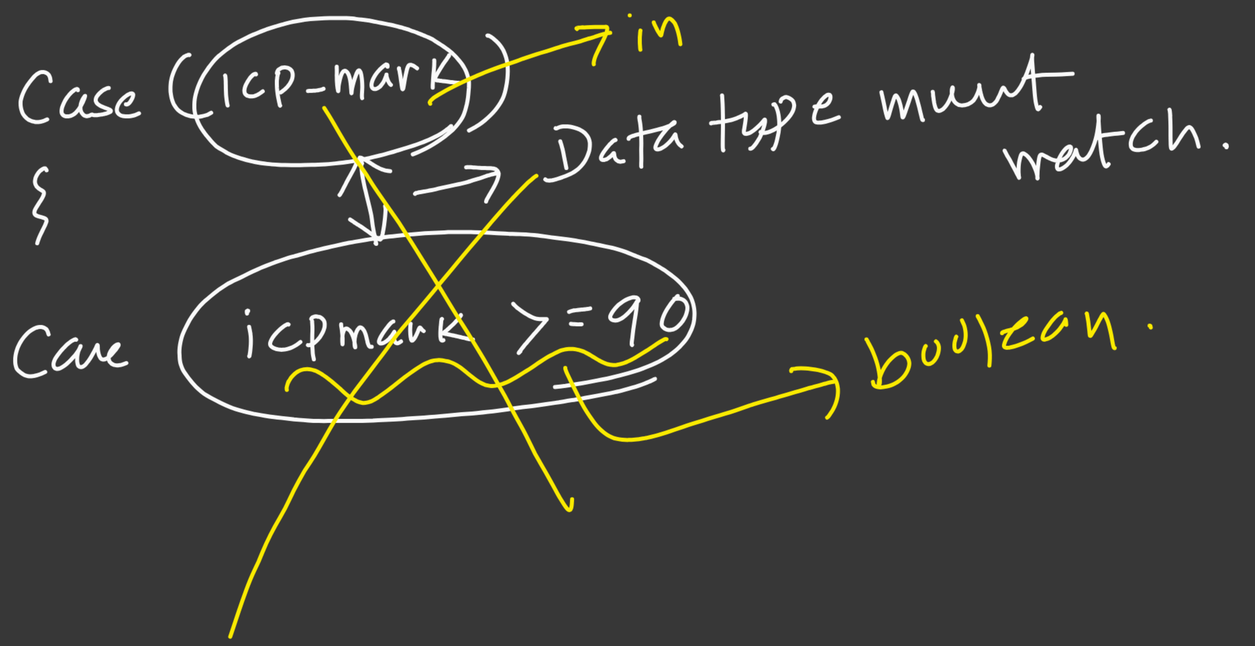
1. Switch Variable
   → The Variable u need to Compare.
2. Case Value.
   → To Set of value that You want to Compare.

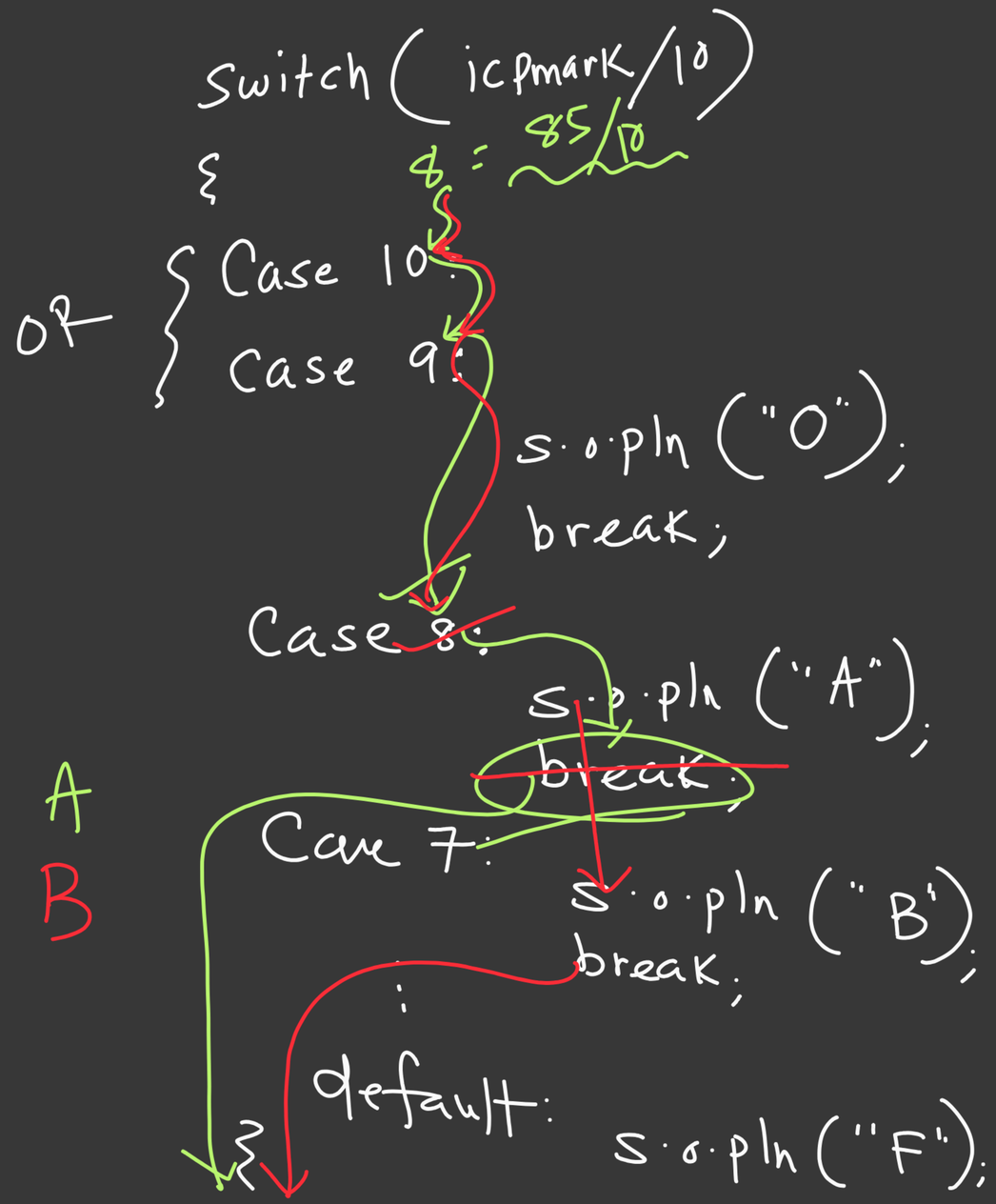


Day

switch Variable Data type & the Case Value Data type must be matched.


dn
1
Monday
Switch(dn)
{
1
2
Case 1:
s.o.pln("Monday");
break;
Case 2:
s.o.pln("Tuesday");
break;
Case 7:
s.o.pln("Sunday");
break;
default:
s.o.pln("Not in Range");
}


Case (icp-mark)
in
Data type must match.
{
Case icpmark >=90
boolean.


Switch (icpmark/10)
{
OR
Case 10:
Case 9:
s.o.pln ("O");
break;
Case 8:
s.o.pln ("A");
break;
A
Case 7:
B
s.o.pln ("B");
break;
default:
s.o.pln ("F");

~~85~~

85

icp. mark	Grade
>= 95	O
85 to 94	A
75 to 84	B
65 to 74	C
55 to 64	D
45 to 54	E
< 45	F

(icp-mark-5)/10

95-5 = 90/10 = 9

99-5 = 94/10 = 9

100-5 = 95/10 = 9

94-5 = 89/10 = 8

85-5 = 80/10 = 8

75-5 = 70/10 = 7

84-5 = 79/10 = 7

```
switch ((icpmark-5)/10)
{
  case 9:
    S.o.pln("O");
    break;

icpmark/5-

Case 19: → O
Case 18 → A
     17
Cas
```

Quiz Test 1

Answer all the questions.

Q.1. Find the limit if it exists.

(a) $\lim_{x \to 0} \frac{|x|}{x}$

(b) $\lim_{x \to 4} f(x)$ where $f(x) = \begin{cases} \sqrt{x-4}, & x > 4 \\ 8 - 2x, & x < 4 \end{cases}$

Q.2. Use implicit differentiation to find an equation of the tangent line to the curve at the given point.

$\sin(x+y) = 2x - 2y$, (π, π)

Q.3. Verify that the given function satisfies the hypotheses of the mean value theorem and find all c that satisfy the conclusion of the theorem.

$f(x) = e^{-2x}$, $[0, 3]$

IPS-1 Quiz.

Q :- write an algorithm, flow chart and execution Pattern or dry Run. to "find Product of individual digit of a given number n"

e.g

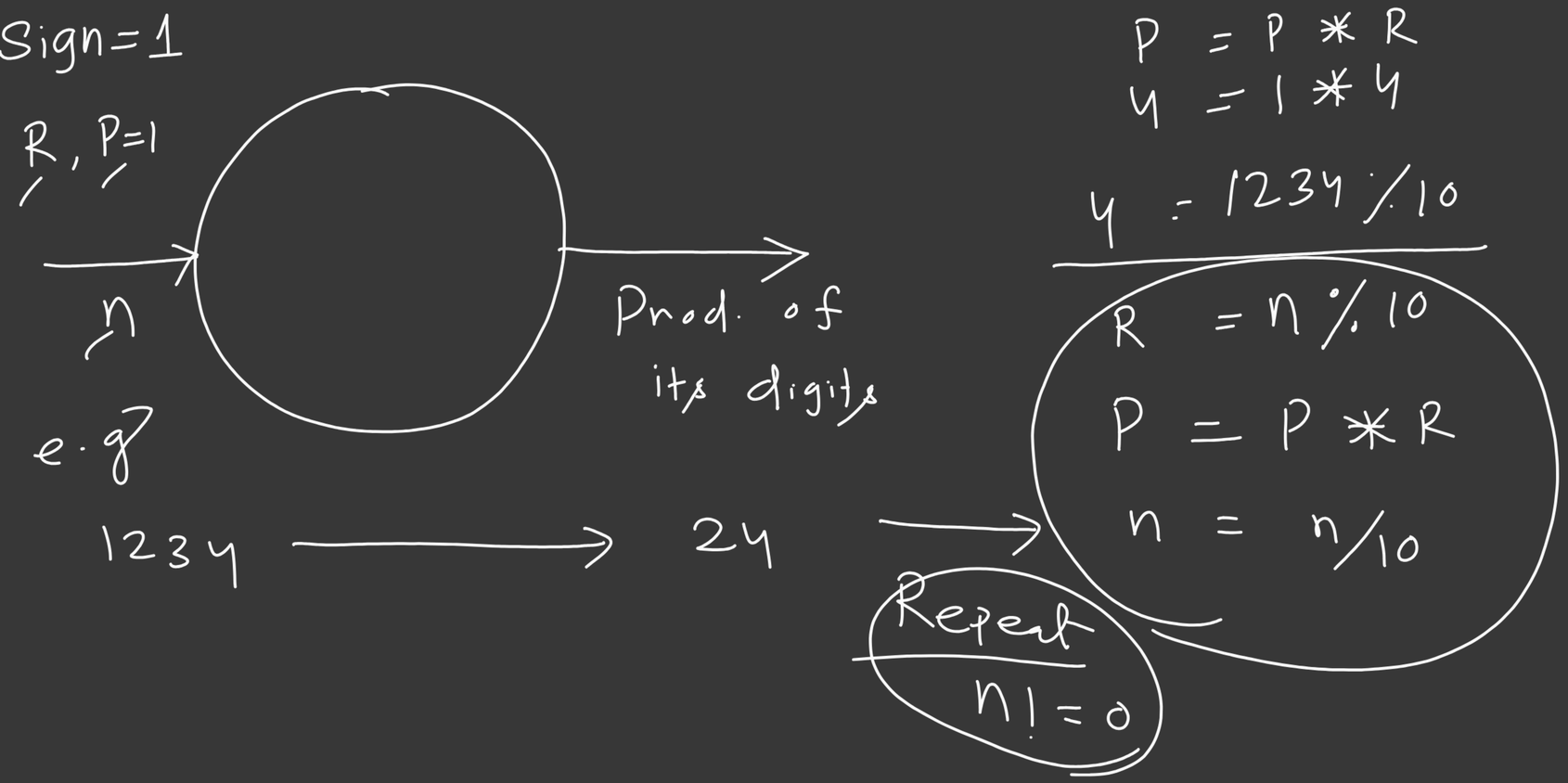
n = 1234 → Product = 24

n = 1201 → Product = 0

n = −125 → Product = −10

n = −1234

Prod = −24




```
if(n<0)
{
  sign = -1;
  n = n * (-1);
}
```

Algorithm

Step 1 :- Input n

step 2 :- R, P = 1, Sign = 1

step 3 :- if (n < 0)

Sign = -1, n = n * (-1)

Step 4 :- Repeat step 5 to 7 until n != 0

Step 5 :- R = n % 10

Step 6 :- P = P * R

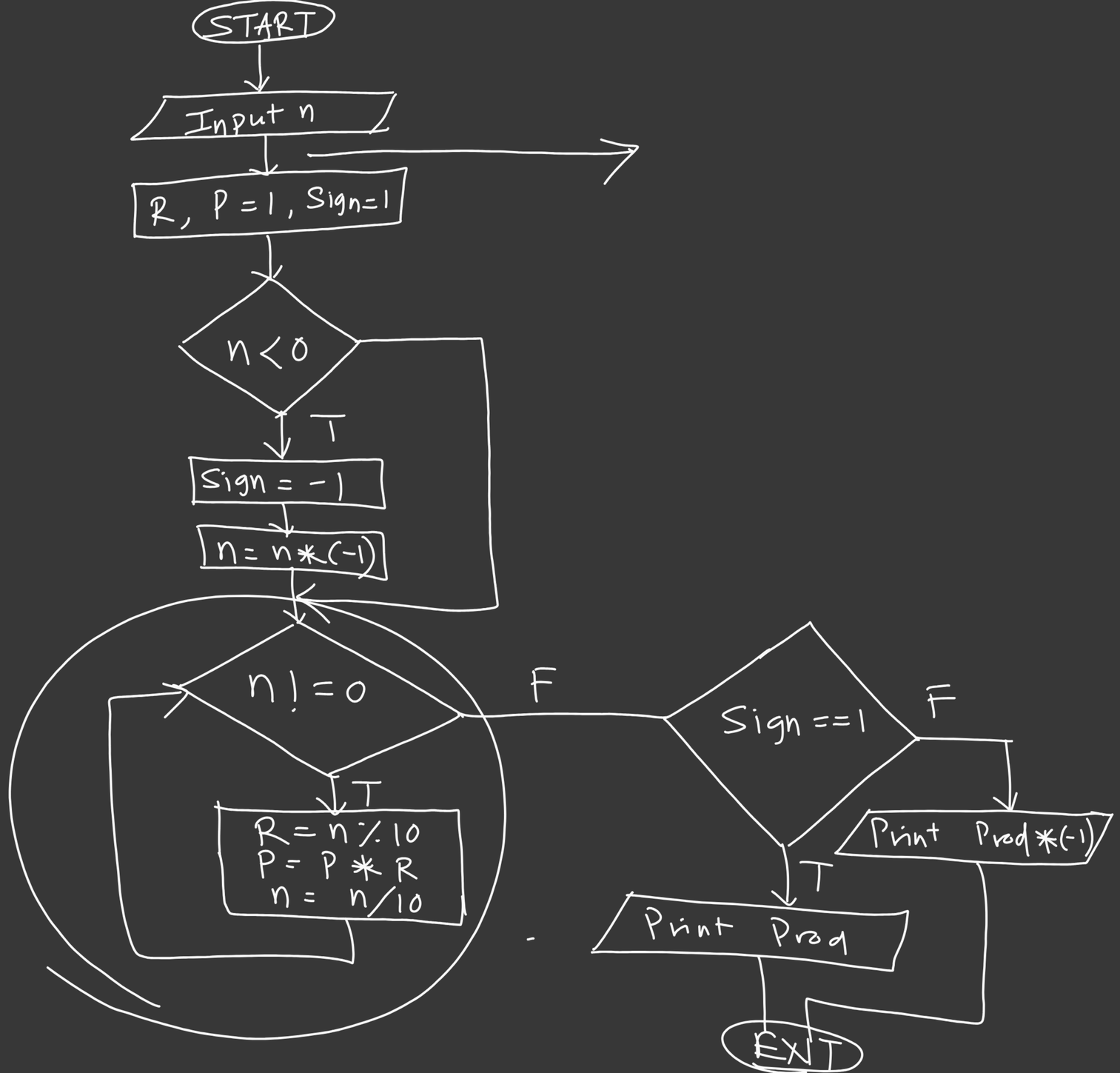
Step 7 :- n = n / 10

Step 8 :- if (Sign == 1)

print Prod

otherwise

print Prod * (-1)


START
Input n
R, P=1, Sign=1
n<0
T
Sign = -1
n = n*(-1)
n!=0
F
T
R = n%10
P = P*R
n = n/10
Sign ==1
F
T
Print Prod*(-1)
Print Prod
EXIT

Ch-3 The fundamentals: Algorithm

Algorithm

Ex: $\{a_1, a_2, a_3, \cdots a_n\}$ find max value ?

Algorithm

```
max := a1
for i = 2 to n
    if max < ai then max := ai
```

i = 2, max < a_2 (T) max := a_2
i = 3, max < a_3 (T) max := a_3
i = 4 max < a_4 (F)
i = 5 max < a_5 (F)
max = a_3 = 8

Ex: $\{1, 5, 8, 3, 6\}$ ($a_1 a_2 a_3 a_4 a_5$)

n = 5, max := a_1

for i = 2 to 5

Searching algorithm

- linear Search
- Binary Search

linear search

find x, $\{a_1, a_2, a_3, \cdots a_n\}$

```
i := 1
while (i ≤ n) & x ≠ ai
    i := i + 1

if (i ≤ n) then location := i
    else location := 0
```

Ex: find x = 7, $\{3, 5, 1, 2, 7, 4\}$ ($a_1 a_2 a_3 a_4 a_5 a_6$)

If n = 6

i = 1

(1 ≤ 6) $x \neq a_1$ (T), i := i + 1 = 2

i = 2

(2 ≤ 6) $x \neq a_2$ (T); i := i + 1 = 3

i = 3

(3 ≤ 6) $x \neq a_3$ (T), i = i + 1 = 4

i = 4, $x \neq a_4$ (T), i = i + 1 = 5

i = 5, $x \neq a_5$ (F)

i = 5

If (5 ≤ 6)(T) location := 5

Ex: find x = 17, $\{3, 1, 13, 10, 7\}$ ($a_1 a_2 a_3 a_4 a_5$)

If n = 5, x = 17

i = 1, (1 ≤ 5)(T) $x \neq a_1$ (T), i = i + 1 = 2

i = 2, (2 ≤ 5)(T) $x \neq a_2$ (T), i = i + 1 = 3

i = 3, (3 ≤ 5)(T) $x \neq a_3$ (T), i = i + 1 = 4

i = 4, (4 ≤ 5)(T) $x \neq a_4$ (T), i = i + 1 = 5

i = 5, (5 ≤ 5)(T), $x \neq a_5$ (T), i = i + 1 = 6

i = 6 (6 ≤ 5)(F) i = 6

If (6 ≤ 5) (F)

location := 0

Binary Search

find

Ex $x = 19$.

$S = \{1, 2, 3, 5, 6, 7, 8, 10, 12, 13, 15, 16, 19, 20, 22\}$

$a_1\ a_2\ a_3\ a_4\ a_5\ a_6\ a_7\ a_8\ a_9\ a_{10}\ a_{11}\ a_{12}\ a_{13}\ a_{14}\ a_{15}$

$n = 15$

$i = 1 \quad j = 15$

$(1 < 15)(T)$, $m = \lfloor \frac{1+15}{2} \rfloor = 8$

If $x > a_8 (T)$ $i := m+1 = 9$

$i = 9, j = 15$

$(9 < 15)$ $m = \lfloor \frac{9+15}{2} \rfloor = 12$

If $x > a_{12} (T)$, $i := 12+1 = 13$

$i = 13, j = 15$

$(13 < 15)(T)$ $m = \lfloor \frac{13+15}{2} \rfloor = 14$

$\{a_1, a_2, \ldots\ldots a_n\}$

$i :=$ . . . .

if $x > a_{14}$ (F)

$j := m = 14$

$i = 13, j = 14$

$(13 < 15)(T)$ $m = \lfloor \frac{13+14}{2} \rfloor = 13$

if $x > a_{13}$ (F)

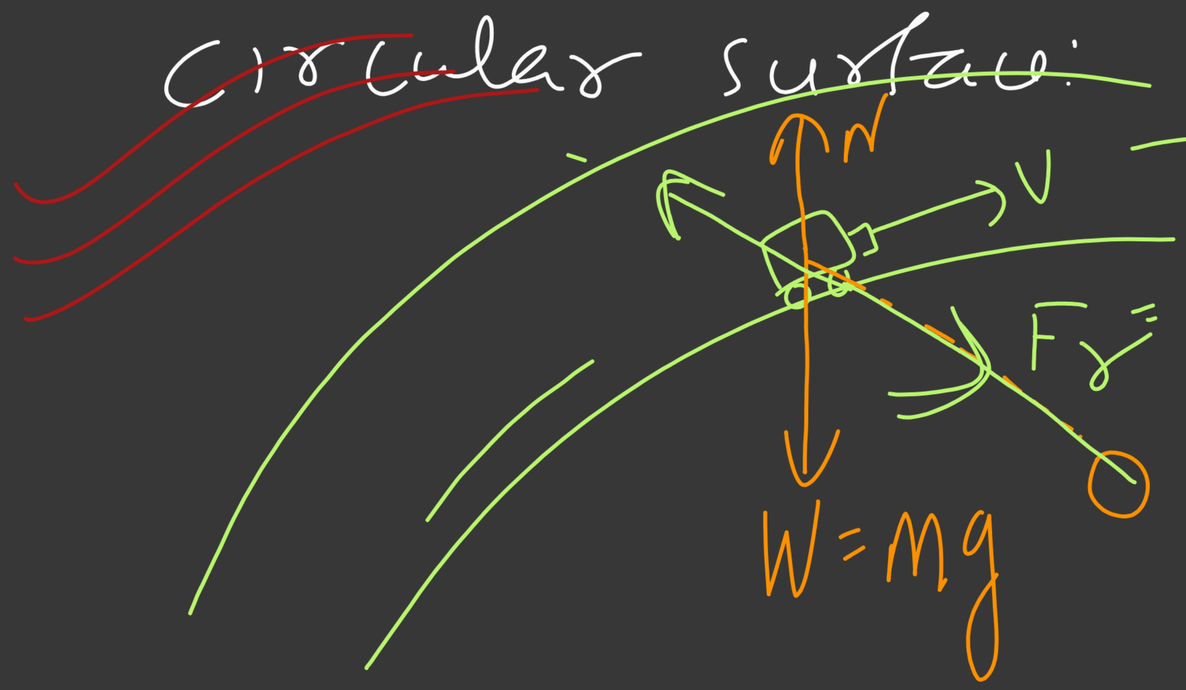
$j = 13$

$i = 13, j = 13$

$(13 < 13)$ (F)

If $x = a_{13} (T)$ location $= 13$

Max^{m} speed of a car on a flat rough circular surface:



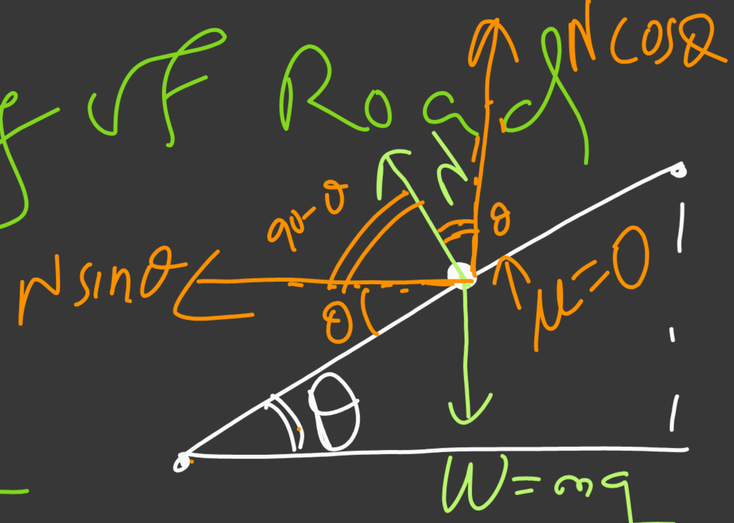
$$N = mg \quad (1)$$

$$F_s = \mu N = \mu mg = \frac{mv^2}{R}$$

$$\Rightarrow v^2 = \mu R g$$

$$\Rightarrow \boxed{v_{max} = \sqrt{\mu R g}}$$

Ex. 5.22 Banking of Road



$mg = N\cos\theta$ — (1)

$\frac{mv^2}{r} = N\sin\theta$ — (2)

$\frac{eq(2)}{eq(1)} \Rightarrow \tan\theta = \frac{v^2}{rg}$

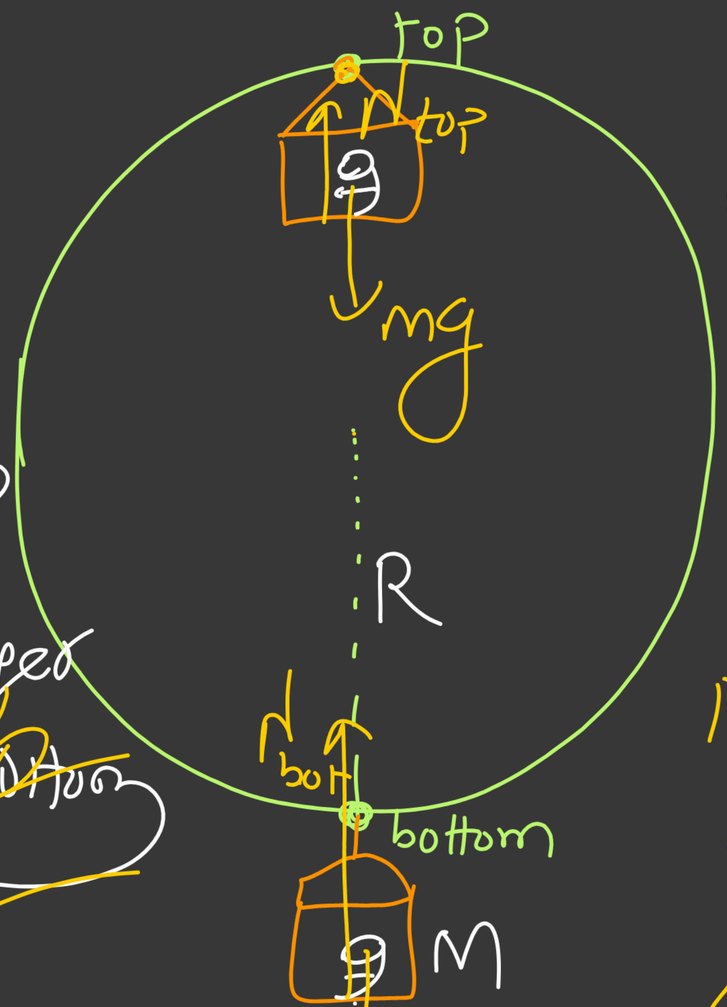
$\Rightarrow v^2 = rg\tan\theta$

$\Rightarrow \boxed{v_{max} = \sqrt{rg\tan\theta}}$

$\sqrt{\mu rg}$

$\boxed{v_{max} = 20\,m/s}$

Ex 5.23



V = const

Q. 5 (C)

Solⁿ

reaction force of seat on passenger at top & bottom

$$N_{bot} - mg = \frac{mv^2}{R}$$

$$\Rightarrow N_{bot} = m\left(g + \frac{v^2}{R}\right)$$

$$mg - N_{top} = mv^2/R$$

$$\Rightarrow N_{top} = mg - \frac{mv^2}{R}$$

$$N_{bot} = mg + mv^2/R$$

$\therefore -v$

TYU S-4

SF. RL. VT

Conclusion about Grav. force b/w Planet & satellite

